STATE OF MATTER

Mention the name of states of matter.

The primary states of matter are- Solid, liquid, gas and plasma. Which distinguised by how there componant particles arranged and behave.

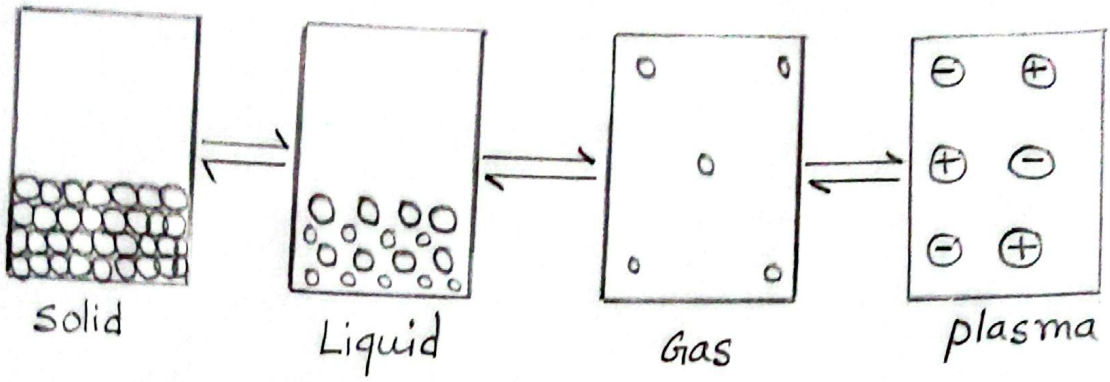
Solid: Tightly packed particles in fixed positions provide a definite shape and volume.

Liquid: Close particles that flow past each other maintain a fixed volume while taking the shape of the container.

Gas: Widely spaced, freely moving particles expand to fill the entire container. The enter atomic distance between the atom is very large.

Plasma: High-energy ionized gas with free electrons influenced by electromgnatic fluids, found naturally in lightning and stars.

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Energy Band in Solids:

The ranges of energy possessed an electron is a solid is known as energy bands.

① Valence band: The range of energyies possessed by valence electrons is known as valence band.

Example:- In case of inert gases, the valence band is full whereas for other materials, it is only partially filled. The partially filled band can accommodate more electrons.

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(11) Conduction band: In certain materials the valence electrons are loosely attached to the nucleus. Even at ordinary temperature, some of the valence electrons may get detached to become free electrons. In fact, it is these free electrons which are responsible for the conduction of current in a conductor. For this reason, they are called conduction electrons. The range of energies possessed by conduction band electrons is known as conduction band. All the electrons in the conduction band are free electrons. Generally, insulators have empty conduction band. On the other hand, it is partially filled for conductors.

Forbidden energy gap:- The seperation between conduction band and valence band on the energy level diagram is known as forbidden energy gap. The width of the forbidden energy gap is a measure of the bondage of valence electrons to the atom.

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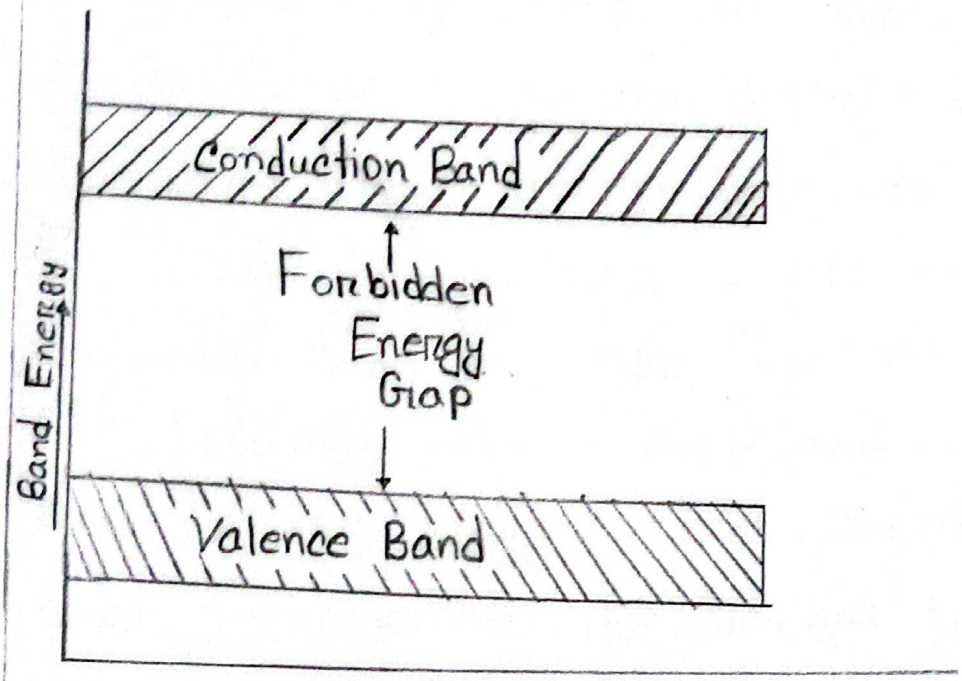


Fig: Energy Band

Classification of Solids and Energy Bands:

The electrons in the lower energy band are tightly bound to the nucleus and play no part in the conduction process. The valence and conduction band are of particular importance in asserttainning the electrical behaviour of various solids.

① Insulators: Insulators are those substances which do not allow the passage of electric current through them. In terms of energy band, the valence band is full while the conduction band is empty. Further, the energy Gap between valence band and conduction band is very large ≃ 15eV. Therefore, at room temperature, the valence band electron of the insulators do not have enough energy to cross over to the conduction band.

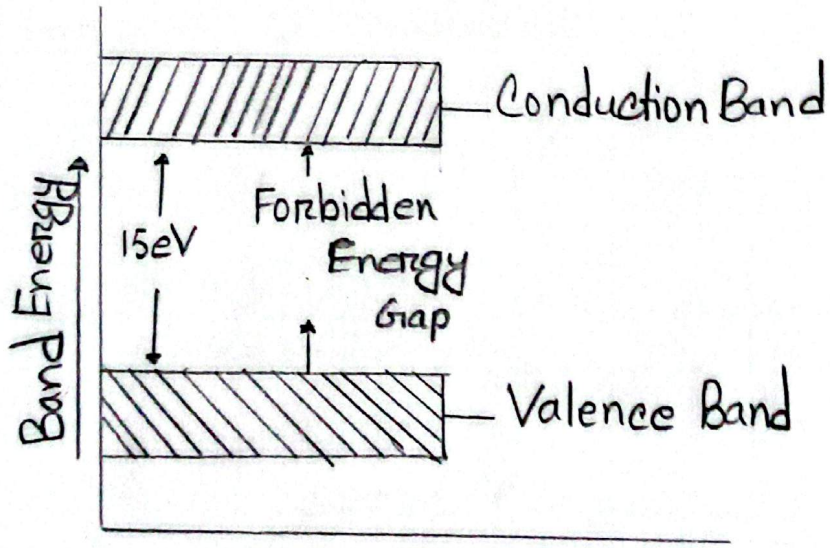


Fig: Insulators

Example: Wood, Glass etc.

(ii) Semiconductors:- semiconductors are those substances whose electrical conductivity lies in between conductors and insulators. In terms of energy band, the valence band is almost filled and conduction band is almost empty. Further, the energy gap between valence and conduction band is very small. Therfore, comparatively smaller electric field is required to push the electrons from the valence band to the conduction band. A semiconductor has:

(a) almost full valence band.

(b) almost empty conduction band.

(c) Small energy gap ($\simeq 1$ eV) between valence and conduction bands.

At low temperature, the valence band is complet full and conduction band is completely empty. Therefore a semeconductor virtually behaves as an insulators at low temperatures. As the temperature is increased, more valence electrons cross over to the conduction band and the conductivity increases.

This shows that electrical conductivity of a semiconductor increases with the rise in temperature.

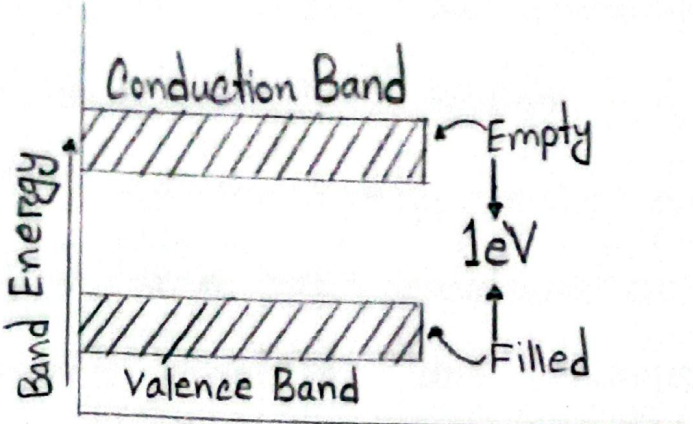


Fig: Semiconductors

Example: Germanium, Silicon.

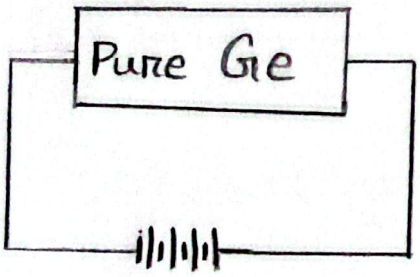
Q: Show that electrical conductivity of a semiconductor increases with the rise in temperature.

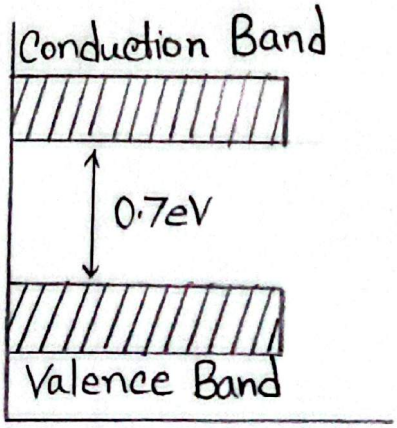
Q: Write the defination insulator and conduction.

Effect of Temperature on Semiconductors:

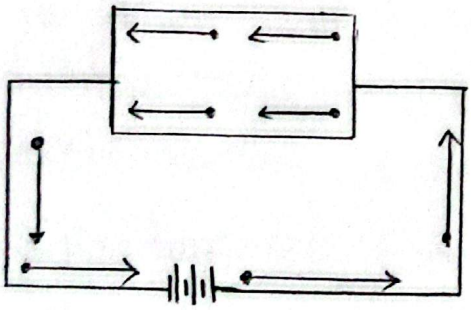
The electrical conductivity of a semconductor change with temperature variations.

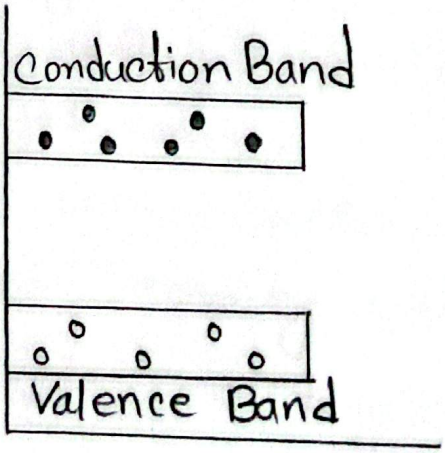
(i) At absolute zero: At absdute zero temperature, all the electrons are tightly held by the semiconductor atoms. The inner orbit electrons are bound whereas the valance electrons are engaged in co-valent binding. At this temperature, the co-valent bonds are very strong and there are no free electrons. so, the semiconductor crystal behaves as a perfect insulator.





(II) Above absolute Zero - When the temperature is raised, some of the co-valent bonds in the semiconductor break due to the thermal energy supplied. The breaking of bonds sets those electrons free, which are engaged in the formation of these bonds. The result is that a few free electrons exist in the semiconductor. These ~~are~~ free electrons can constitute a tiny electric current if potential difference is applied across the semiconductor. This shows that the resistance of a semiconductor decreases with the rise in temperature.





Hole Current: When co-valent bond is broken due to thermal energy, the removal of one electron is vacancy, a missing electron in the co-valent band. These missing electron is called a hole. Which acts a positive charge.

Classification of Semiconductor:

(I) **Intrinsic Semiconductor:-** A semiconductor is an extremly pure form is known as intrinsic semiconductor.

(II) **Extrinsic Semiconductor:-** When a small amount suitable impurity is added to the semiconductor It is then called impurity or extrinsic semiconductor. The process of adding impuritigs to a semiconductor is known as doping.

Classification of Extrinsic semiconductor:-

Depending upon the type of impurity added. Extrinsic semiconductor classify into two types-

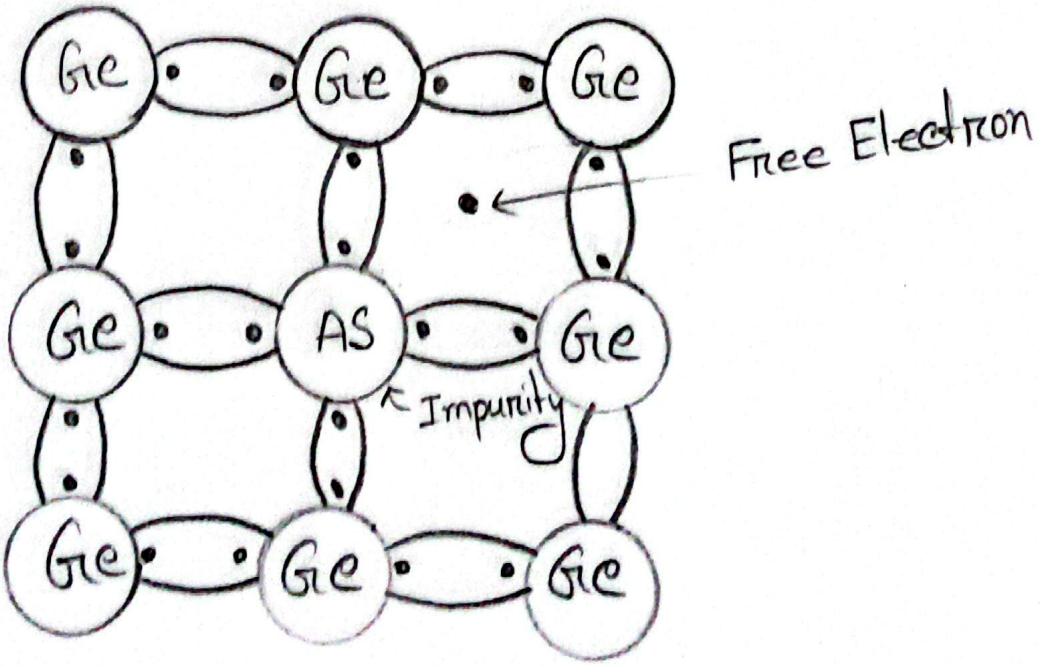
(I) n-type Semiconductor.

(II) P-type Semiconductor.

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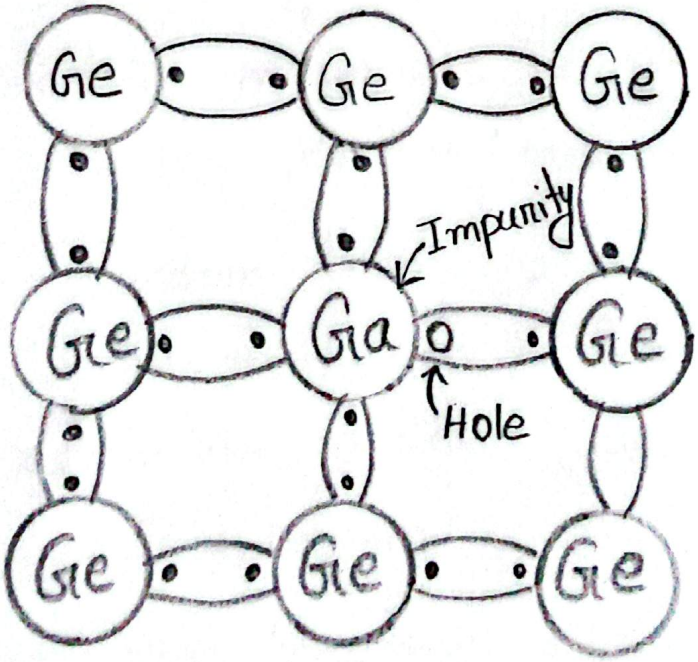
n-type Semiconductor:-

When a small amount of pentavalent impurity is added to a pure semiconductor, it is known as n-type semiconductor. The addition of pentavalent impurity provides a large amount of free electrons in the semiconductor crystal. Typical example of Pentavalent impurities are Arsenic (atom 33) and Antimony (atom 51).

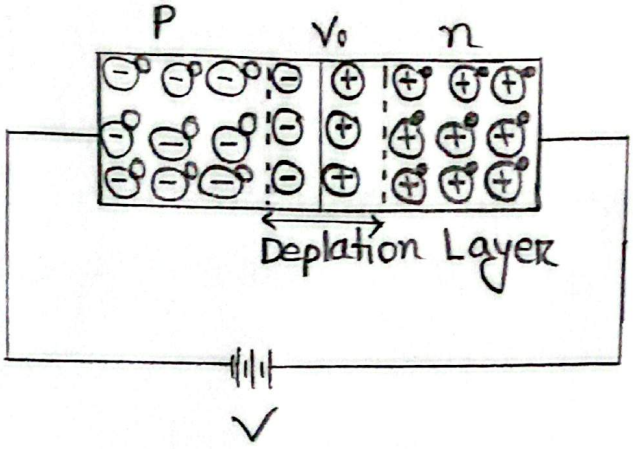


P-type Semiconductor:

When a small amount of trivalent impurity is added to a pure semiconductor, it is known as P-type semiconductor. The addition of trivalent impurity provides a large amount of holes in the semiconductor. Typical examples of trivalent impurities are Gallium (Atom 31) and Indium (atom 49).



P.n-Junction: When a p-type semiconductor is suitably joined to a n-type semiconductor. The contacts surface is called p.n junction.



In this figure left side material p-type semiconductor having negative ions and positive holes. The right side materials is n-type semiconductor haring positive ions and free electro

1. At the junction three is a tendency for the free electrons to diffuse over to p-side and holes to the n-side.. This process is called diffusion. This region is called deplation.
2. As the free electron moves from n-type to p-type positive ions are uncovered. Hence, a positive charge is build on the n-side of the junction.

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3. At the same time the free electrons cross the junction by filling in the holes. Therefore, a net negative charge is established on the p-side of the junction.

4. Thus a barrier is set up against the movement of charge carrier. This is called potential barrier.

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